

Importance of N₂H_x Reaction Pathways in Modelling of NH₃/H₂/air Flame Chemistry

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ABSTRACT

A reduced kinetic mechanism was optimized for NH₃, NH₃/CH₄, and NH₃/H₂ flames to support efficient CFD modeling. 1D flame analyses showed that the under-prediction of burning velocity in mixtures with high-hydrogen blends by the original mechanism stem from insufficient H-atom generation via $\text{NH}_2 + \text{NH} \rightleftharpoons \text{N}_2\text{H}_2 + \text{H}$ and $\text{NNH} \rightleftharpoons \text{N}_2 + \text{H}$. Updating N₂H_x pathways resolved this issue, yielding accurate burning velocity predictions across all target blends.

1. Introduction

Ammonia (NH₃) is a promising carbon-free fuel, frequently blended with methane (CH₄) or hydrogen (H₂) to enhance reactivity in combustion systems. Combustor development for these blends relies on numerical simulations to accurately predict key phenomena, including laminar burning velocity, ignition dynamics, and pollutant emissions. Because large-scale or three-dimensional modeling using detailed kinetics incurs substantial memory and computation time costs, reliable reduced mechanisms, containing fewer species and reactions, are essential.

Numerous chemical kinetic mechanisms have been developed to model ammonia combustion; however, most are optimized for specific fuel blends and narrow operating conditions. For example, the reduced mechanism by Okafor et al. [1] was designed for NH₃/air and NH₃/CH₄/air flames. While it effectively predicts unstretched laminar burning velocity (S_L) across lean-to-rich equivalence ratios for these mixtures [1–3], it significantly underpredicts S_L for flames when H₂ volume fractions exceed 0.3. Conversely, the detailed and reduced mechanisms by Gotama et al. [4] were tailored for NH₃/H₂/air combustion, accurately predicting S_L for high-hydrogen blends and pure H₂/air flames at atmospheric pressure [4]. Similarly, the mechanism by Nakamura et al. [5] captures S_L well for fuel-rich NH₃/H₂/air flames ($\phi = 1.1 - 1.4$) [5]. Consequently, there remains a critical need for a single reduced mechanism capable of accurately evaluating the unstretched laminar burning velocity across a wide spectrum of ammonia blends and operating conditions.

Okafor et al.'s reduced mechanism [1] under-prediction of S_L for NH₃/H₂/air flames could lie in the complex cross-coupling between ammonia and hydrogen reaction pathways. Because this reduced mechanism accurately models NH₃/air, NH₃/CH₄/air and H₂/air flames on their own, its failure at higher hydrogen fractions ($X_{\text{H}_2} > 0.3$) points to missing or miscalibrated kinetic interactions specific to NH₃/H₂ co-combustion. At elevated hydrogen levels, the surge in key radical pools (H, O, and OH) alters the primary oxidation routes and radical consumption rates that dictate flame speed [4,5]. These limitations suggest that the mechanism's reaction network may need to be re-examined specifically under conditions where H₂ concentration in the fuel mixture is significant.

The primary objective of this work is to develop a

robust reduced chemical kinetic mechanism capable of accurately modelling the combustion of NH₃/air, NH₃/CH₄/air, and NH₃/H₂/air fuel blends across a wide range of operating conditions, building directly upon the reduced model of Okafor et al. [1]. To achieve this, the underlying reaction pathways of NH₃/H₂/air flames were analysed to elucidate key kinetic interactions between ammonia and hydrogen chemistries that were underrepresented in the original mechanism.

2. Numerical Procedure

In this study, one-dimensional unstretched adiabatic simulations were conducted to evaluate the unstretched laminar burning velocity, S_L , of the premixed flames using the Premixed Laminar Flame Speed Calculation Model in ANSYS Chemkin-PRO. The computation domain was set 10 cm and S_L was calculated at the cold boundary of the computation domain. The adaptive grid settings GRAD and CURV were both set to 0.01. The ratio of H₂ in the binary fuel, X_{H_2} , is defined in Equation (1). X_{H_2} was varied, while atmospheric pressure and stoichiometric equivalence ratio ($\phi = 1.0$) were maintained, to study how each reaction mechanism's S_L predictions are affected by hydrogen addition.

$$X_{\text{H}_2} = \frac{[\text{H}_2]}{[\text{NH}_3] + [\text{H}_2]} \quad (1)$$

The software was also used to compute rate of production of species, species concentrations and sensitivity of S_L to rate constants of elementary reactions. The reduced chemical kinetics mechanism by Okafor et al. [1] and Gotama et al.'s [4] detailed mechanism was used in the simulations. Henceforth, for the purpose of discussion, these mechanisms shall be referred to as Okafor Reduced Mech and Gotama Mech respectively. A reduced chemical kinetic model, based on Okafor Reduced Mech [1], with 43 species and 152 reactions was developed as described in the following section. The developed mechanism will be referred to as Present Mech in this paper.

3. Development of the Present Mech

The new reaction mechanism developed in this paper was based on the Okafor Reduced mechanism. Okafor Reduced Mech was selected because this mechanism has been optimized and validated for S_L prediction of NH_3/air and $\text{NH}_3/\text{CH}_4/\text{air}$ flames. This is true for varying equivalence ratios, pressures and, in the case of $\text{NH}_3/\text{CH}_4/\text{air}$ flames, varying methane concentrations. This mechanism also models the S_L of H_2/air flames satisfactorily for most of these conditions. However, it significantly under-predicts S_L of $\text{NH}_3/\text{H}_2/\text{air}$ flames with hydrogen fractions larger than 0.3 as shown in Fig. 1, hence motivating the need to optimize it for $\text{NH}_3/\text{H}_2/\text{air}$ flames.

Figure 1 highlights Gotama Mech's S_L predictions agree with experimental measurements of S_L across all X_{H_2} . Understanding where Gotama Mech differs from Okafor Reduced Mech should provide insights as to where the under-prediction of $\text{NH}_3/\text{H}_2/\text{air}$ flames stems from. In addition, development of Gotama Mech focused on improving $\text{NH}_3/\text{H}_2/\text{air}$ flame speed prediction at intermediate hydrogen concentration [4].

To understand how oxidation kinetics of each fuel in the binary fuel interact with each other, reaction pathways for each mechanism were constructed using integrated rate of production (IROP) of major species related to NH_3 and H_2 oxidation. Each IROP is calculated by integrating the rate of production, $\dot{\omega}_{R,i}$, of specie i in elementary reaction R along the length of the computational domain (10 cm) using Equation (2). The IROP, hence,

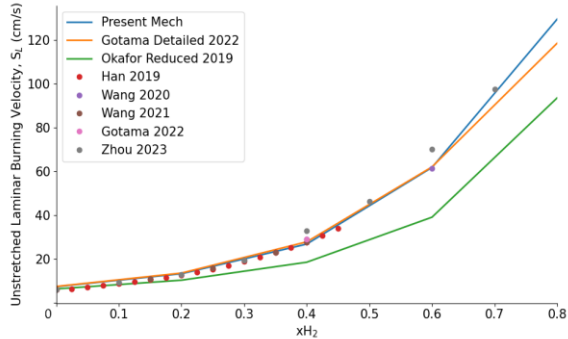


Fig. 1 Experimental and numerical laminar burning velocity of $\text{NH}_3/\text{H}_2/\text{air}$ flames under stoichiometric condition at various hydrogen concentrations.

characterizes the contribution of each elementary reaction to the combustion of $\text{NH}_3/\text{H}_2/\text{air}$ fuel in each chemical kinetics model.

$$\int_0^L \dot{\omega}_{R,i} dx \quad (2)$$

Figures 2 (a) and (b) are reaction pathways of stoichiometric $\text{NH}_3/\text{H}_2/\text{air}$ flame of $X_{\text{H}_2} = 0.5$ computed using Okafor Reduced Mech and Gotama Mech respectively. The thickness of each arrow corresponds to the magnitude of each path's IROP. Species besides each arrow have numbers that represent

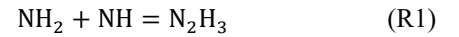
the percent contribution each elementary reaction has to the corresponding path. The pathways highlighted the key differences in how the interaction of NH_3 , and H_2 oxidation paths are modelled in each mechanism to inform the development of Present Mech.

Normalized sensitivity of S_L to rate constants of elementary reactions was subsequently analysed to determine the direct dependence of the laminar burning velocity (S_L) on individual elementary reactions. By combining these sensitivity trends with IROP analyses, the dominant reactions and critical cross-coupling pathways dictating flame speed were successfully identified. Based on these insights, targeted modifications made to Okafor Mech to model the under-represented pathways for $\text{NH}_3/\text{H}_2/\text{air}$ flames at higher hydrogen fractions and develop the Present Mech.

Present Mech then was validated using experimental flame speed measurements from literature for $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CH}_4/\text{air}$ flames across lean ($\phi = 0.7$), stoichiometric and rich ($\phi = 1.4$) equivalence ratios. Present Mech's performance at elevated pressures were evaluated too. These validation steps ensure that the developed mechanism meets the aim of this study and is compatible with various ammonia fuel blends at various operating conditions.

4. Results and Discussion

Comparisons between the pathways of Okafor Reduced Mech and Gotama Mech, Fig. 2 (a) and (b) respectively, highlight interactions between NH_3 and H_2 oxidation pathways at intermediate H_2 concentrations in the binary fuel are modelled differently in the two chemical kinetics models. For instance, reactions in the reaction path $\text{NH}_2 \leftrightarrow \text{N}_2\text{H}_3$ is modelled in Gotama Mech only. The forward reaction, the formation of N_2H_3 , is given by;



and the reverse, the consumption of N_2H_3 , is given by;



Gotama Mech's reaction pathway depicts the involvement of N_2H_3 in the control of intermediates important in ammonia oxidation pathway.

The reaction pathway $\text{NNH} \rightarrow \text{N}_2$ are distinct between Okafor Reduced Mech and Gotama Mech. Figure 2 shows that Gotama Mech exhibits an IROP approximately 2.2 times larger than Okafor Reduced Mech and the pathway is primarily dominated by the reaction;



in Gotama Mech, which contributes significantly to H radical generation at intermediate hydrogen concentration.

Another reaction pathway, $\text{NH}_2 \rightarrow \text{N}_2\text{H}_2$ can also be depicted to exhibit increased IROP in Gotama Mech, approximately 3 times larger than in Okafor Reduced Mech. The reaction;

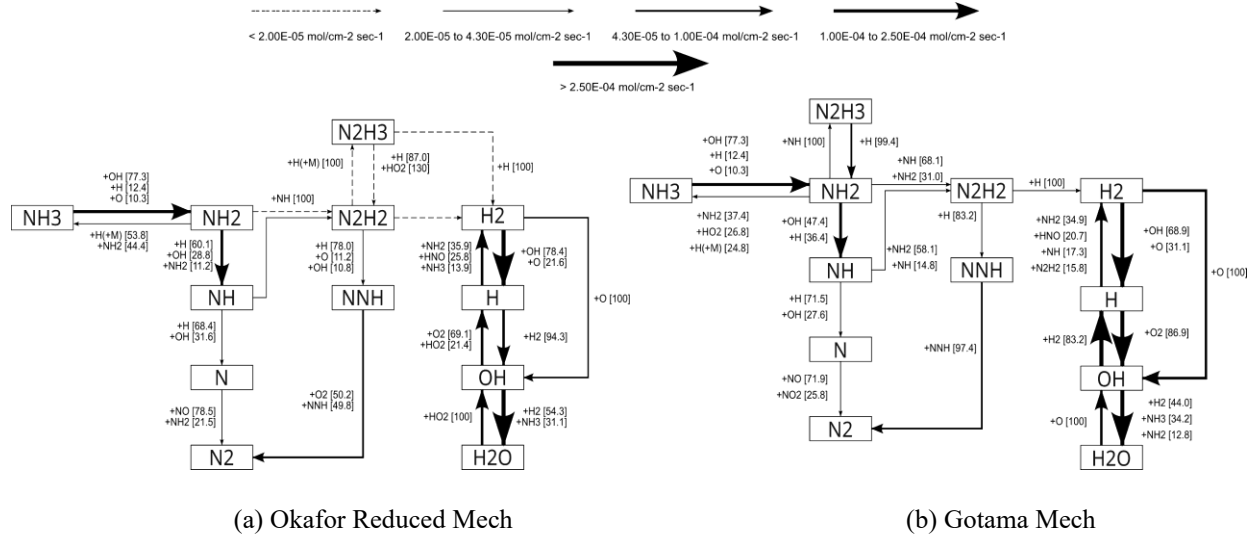
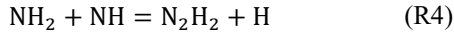


Fig. 2 Reaction pathway of $\text{NH}_3/\text{H}_2/\text{air}$ flame at $X_{\text{H}_2}=0.5$ and equivalence ratio of 1.0 using (a) Okafor Reduced Mech and (b) Gotama Mech.



in the biggest contributor to the pathway. (R4) is a H radical generating reaction while consuming NH_2 and NH intermediates. Moreover, the increased N_2H_2 production further contributes to promoting the pathways $\text{N}_2\text{H}_2 \rightarrow \text{NNH} \rightarrow \text{N}_2$ as the IROP for these pathways in Gotama Mech were approximately 2.8 times and 2.2 times that of Okafor Reduced Mech respectively. In addition, (R1) is the significant reaction in the latter pathway suggesting.

Reactions (R1) and (R2) were added to Present Mech as these reactions were established to be important for controlling NH concentration. These reactions' roles in the combustion of $\text{NH}_3/\text{H}_2/\text{air}$ flames with intermediate hydrogen concentrations were also confirmed by Gotama et al. [4]. The rate constants of (R3) was updated by choosing rate constants published by Pagsberg et al. [6]. This enhanced the production of H atoms from the $\text{NNH} \rightarrow \text{N}_2$ pathway. Glarborg et al. [7] corroborates the dominance of reaction (R3) in this pathway and confirms the significant role (R3) plays in H radical generation. Rate constants for (R4) were updated with the rate constants published by Klippenstein et al. [8]. Otomo et al. also recognizes the influence of reaction (R4) on the S_L of $\text{NH}_3/\text{H}_2/\text{air}$ flames.

Normalized sensitivity of S_L to rate constants of the elementary reactions using the Present Mech and Okafor Mech highlighted reaction (R4) playing a significant role in promoting S_L in Present Mech while remaining insignificant in Okafor Reduced Mech. In addition, reaction (R1) only significantly inhibits S_L in the Present Mech while remaining insignificant for Okafor Reduced Mech. This analysis also showed other NH_x and N_2H_x related reactions being more significant with respect to S_L in Present Mech at intermediate hydrogen concentration.

The reactions (R2), (R3) and (R4) involve N_2H_x

species and control H radical concentrations, an important radical for promoting S_L . The increased IROP of these reactions in Present Mech implies an increase in H radical concentration, which correlates with the increased burning velocity predictions of this mechanism relative to Okafor Reduced Mech. Moreover, the formation and consumption of N_2H_3 with (R1) and (R2) respectively control NH and NH_2 concentrations. NH_x species are important as they interact with H and OH radicals and produce these radicals. (R4) is an example of NH_x species' involvement in the production of an H radical. The other product of (R4), N_2H_2 , is part of the pathway to form NNH which also produces H radicals via (R3). Moreover, reaction (R4)'s significant positive effect on the burning velocity further supports the theory that N_2H_x species and their reaction pathways are important to accurately describe flame with fuel compositions of intermediate hydrogen concentration. Therefore, the Present Mech with a more accurate N_2H_x chemistry can predict S_L under these conditions accurately.

5. Validation

The Present Mech was validated with experimental data for S_L of $\text{NH}_3/\text{H}_2/\text{air}$ from the datasets consolidated by Han et al. [9]. These data sets were proven to be consistent with each other and widely used chemical kinetic mechanisms. Present Mech was validated for $\text{NH}_3/\text{CH}_4/\text{air}$ flames also. Experimental data for $\text{NH}_3/\text{CH}_4/\text{air}$ S_L was compiled from three datasets: Okafor et al. (2019) [1], Han et al. (2019) [10] and Ling et al. (2026) [11]. However, experimental datasets, for both fuel blends, lacked ample S_L measurements at elevated pressures, making validation of Present Mech for high pressure combustion difficult.

Figure 3 compares $\text{NH}_3/\text{CH}_4/\text{air}$ predictions by the Present Mech to experimentally measured S_L from literature at stoichiometric equivalence ratio and

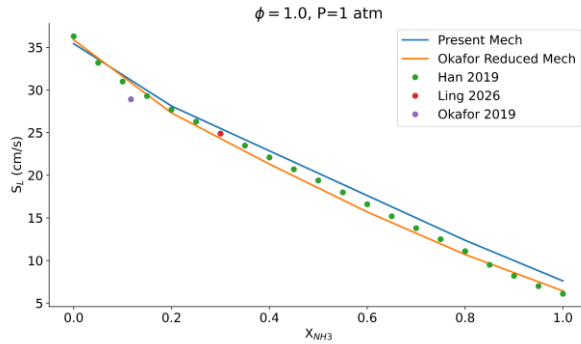


Fig. 3 Comparison of measured and simulated laminar burning velocity of $\text{NH}_3/\text{CH}_4/\text{air}$ flames under stoichiometric condition and 1 atm of pressure.

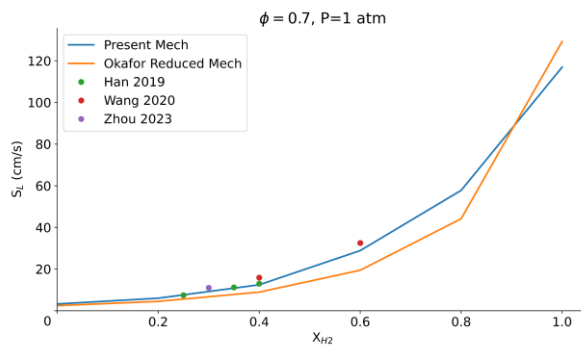


Fig. 4 Comparison of measured and simulated laminar burning velocity of $\text{NH}_3/\text{H}_2/\text{air}$ flames under lean condition ($\phi = 0.7$) and 1 atm of pressure.

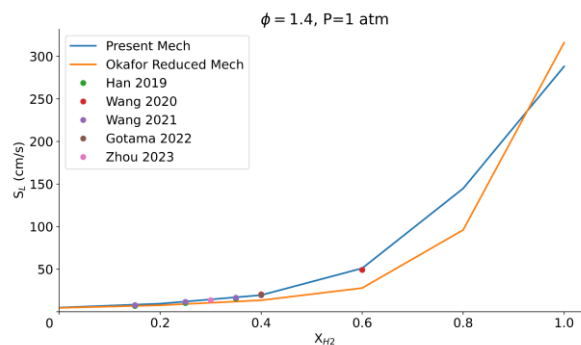


Fig. 5 Comparison of measured and simulated laminar burning velocity of $\text{NH}_3/\text{H}_2/\text{air}$ flames under lean condition ($\phi = 1.4$) and 1 atm of pressure.

atmospheric pressure to depict that the S_L predictions are satisfactory. This shows that the present optimization of the N_2H_x sub-mechanism does not undermine the fidelity of the mechanism in modelling $\text{NH}_3/\text{CH}_4/\text{air}$ blends, which the original mechanism was optimized for.

Figure 4 and 5 shows that the Present Mech satisfactorily models the laminar burning velocity of $\text{NH}_3/\text{H}_2/\text{air}$ at lean and rich equivalence ratios respectively. The new reduced model was also validated with measurements from Wang et al. [12] at 3 atm and

intermediate H_2 concentrations ($X_{\text{H}_2} = 0.4, 0.6$) and Gotama et al. [4] at 5 atm and $X_{\text{H}_2} = 0.4$. This suggests that Present Mech is now capable of modelling flame speed of hydrogen enriched ammonia combustion.

6. Conclusion

In this study, a reduced chemical kinetics model was optimized for NH_3/air , $\text{NH}_3/\text{CH}_4/\text{air}$ and $\text{NH}_3/\text{H}_2/\text{air}$ flames, based on an existing chemical kinetics model developed for NH_3/air and $\text{NH}_3/\text{CH}_4/\text{air}$ flames. A comparative chemical kinetics analysis of the original reduced mechanism to detailed reaction mechanisms in literature showed that the significant underprediction of the laminar burning velocity of the $\text{NH}_3/\text{H}_2/\text{air}$ flames at intermediate hydrogen volume fractions was due to inaccurate modelling of H atom production from the N_2H_x sub-mechanism. By incorporating elementary reactions and optimizing key rate constants within the N_2H_x sub-mechanism, $\text{NH}_3/\text{H}_2/\text{air}$ laminar burning velocity predictions were made more accurate without compromising the mechanism's fidelity for NH_3/air and $\text{NH}_3/\text{CH}_4/\text{air}$ flames. This study therefore shows the unique importance of the N_2H_x sub-mechanism in modelling the flame speeds of $\text{NH}_3/\text{H}_2/\text{air}$ flames.

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